Business Models Unlocked by wasm4pm

A Receipt-Native Process Mining Platform
Compiled to WebAssembly

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Abstract

wasm4pm is a process mining platform with 60 deterministic algorithms compiled from Rust to WebAssembly, wrapped in a TypeScript monorepo, and governed by a BLAKE3 receipt chain that makes every analytical result cryptographically attributable to its inputs. This document analyzes the business models that this architecture unlocks: dual licensing under BUSL-1.1, evidence-as-a-service for regulated industries, edge and embedded deployment economics, and an ecosystem strategy built on verifiable computation. The thesis throughout is that the unit of value is not the algorithm — algorithms are commodities — but the *receipt*: a portable, verifiable proof that a specific analysis ran on specific data and produced a specific result.

Contents

1 The Platform as an Economic Object **1**
1.1 What wasm4pm actually is . . . 1
1.2 Determinism as a product feature . . . 1
1.3 The receipt chain . . . 1
1.4 WebAssembly as a distribution strategy . . . 1
1.5 The unit of value . . . 2

2 The Dual-License Engine: BUSL-1.1 Plus Commercial **3**
2.1 The structure . . . 3
2.2 Why BUSL fits this product specifically . . . 3
2.3 Pricing axes . . . 3
2.4 The CalVer signal . . . 4

3 Evidence-as-a-Service **5**
3.1 The category . . . 5
3.2 The mechanism . . . 5
3.3 Who pays, and under which mandate . . . 6
3.4 Why incumbents cannot trivially follow . . . 6
3.5 Revenue shape . . . 6

4 Edge Economics: Where WASM Changes the Cost Curve **7**
4.1 The incumbent cost structure . . . 7
4.2 The wasm4pm inversion . . . 7
4.3 Business models this enables . . . 7
4.4 The size gate as a commercial control point . . . 8

5 Vertical Plays **9**
5.1 CI/CD process assurance (the proven test case) . . . 9
5.2 Healthcare pathway conformance . . . 9
5.3 Financial transaction monitoring . . . 9
5.4 Manufacturing and Industry 4.0 . . . 9
5.5 Sequencing . . . 10

6 Ecosystem and Platform Strategy **11**
6.1 The package surface as a funnel . . . 11
6.2 The agent and MCP channel . . . 11
6.3 Marketplace dynamics . . . 11
6.4 The standards position . . . 12

7 Risks, Honest Limitations, and Roadmap **13**
7.1 Risks stated plainly 13
7.2 Sequenced roadmap 13
7.3 Closing argument 14

Chapter 1

The Platform as an Economic Object

1.1 What wasm4pm actually is

wasm4pm is two layers with distinct economic properties:

1. A **Rust/WASM core** containing 60 registered process mining algorithms — discovery (DFG, heuristic, inductive), conformance checking, performance analysis, and object-centric querying — compiled to WebAssembly. Deterministic by construction: the same input produces a bit-exact output, with all randomness seeded and all hash-map iteration sorted.

2. A **TypeScript monorepo** of eleven packages that wrap the core with a professional CLI (`wpm`), Zod-validated configuration, an execution planner with four deployment profiles, OpenTelemetry observability, and a Supabase persistence layer.

Three architectural decisions create the commercial surface area this document explores.

1.2 Determinism as a product feature

Most analytics platforms treat reproducibility as a best effort. wasm4pm treats it as a merge gate: same input, bit-exact output, enforced in CI. Commercially, determinism converts an analytical claim into an *auditable artifact*. A compliance officer does not have to trust that the conformance score was 0.94; they can re-run the analysis and obtain the identical bytes. Determinism is the precondition for everything in Chapter 3.

1.3 The receipt chain

Every `wpm run` emits a BLAKE3 receipt to `.wasm4pm/receipts/` containing the run identifier, the command, a hash of every input, a hash of every output, and the execution status. The receipts compose into a chain: the output hash of one analysis becomes the input hash of the next. The chain is the platform's most defensible asset because it is *accumulated state* — a competitor can reimplement an algorithm in a weekend, but cannot reproduce a customer's two-year evidence history.

1.4 WebAssembly as a distribution strategy

Compiling to WASM collapses the deployment matrix. One artifact runs in the browser, in Node.js, on edge runtimes (Cloudflare Workers, Fastly), and inside any host language with a WASM runtime (Python, Go, JVM, .NET). The size-profiled builds — mobile (∼500 KB), IoT (∼1 MB), edge (∼1.5 MB), fog (∼2 MB), and full browser (∼3.5 MB) — mean the same algorithm catalog can be sold into radically different deployment contexts without porting cost. Chapter 4 prices this.

1.5 The unit of value

The framing assumption for every model that follows:

> *Algorithms are commodities; pm4py gives away a superset of this catalog for free. The sellable unit is the receipt — the verifiable, portable, machine-checkable proof that a specific analysis ran on specific data and produced a specific result, on infrastructure the customer controls.*

Chapter 2

The Dual-License Engine: BUSL-1.1 Plus Commercial

2.1 The structure

wasm4pm is licensed under the Business Source License 1.1 with a commercial license available for production use that falls outside the BUSL grant. After the Change Date the code converts automatically to AGPL-3.0. This is the MariaDB/HashiCorp/Sentry pattern, and it creates a three-segment funnel:

Free tier (BUSL-compliant use). Evaluation, development, internal tooling, academic research. This population is the adoption engine — it produces GitHub stars, npm downloads, bug reports, and the analysts who later champion a paid deployment.

Commercial tier. Anyone embedding wasm4pm in a product or service offered to third parties, or whose production use exceeds the BUSL additional-use grant. This is the revenue engine.

Post-Change-Date tier. Code older than the change window is AGPL — a strong-copyleft poison pill for proprietary embedders that keeps the commercial license attractive even for old versions.

2.2 Why BUSL fits this product specifically

BUSL is usually defended as cloud-provider insurance. For wasm4pm there is a sharper argument: **the product is a trust primitive, and trust primitives must be source-available.** A receipt chain whose verifier is closed-source is marketing, not evidence. A regulated customer's auditors must be able to read the BLAKE3 hashing path and the conformance scoring code. Fully proprietary licensing would destroy the core value proposition; fully permissive licensing (the previous MIT/Apache posture) would invite a hyperscaler to operate "Process Mining as a Service" on the project's own code. BUSL is the narrow corridor between those failure modes.

2.3 Pricing axes

The architecture suggests charging on axes that scale with value received, not with seats:

Axis	Rationale
Receipts retained / verified per month	Directly proportional to audit value; metered by the Supabase sync layer that already counts them.
Deployment profiles unlocked	Browser-only free; edge/IoT/fog profiles commercial — the size-optimized builds are real engineering, easily gated.
Algorithm tiers	Core discovery free; conformance, OCEL 2.0 querying, and the cognition layer commercial.
Embedding rights	OEM license for ISVs shipping wasm4pm inside their own products — the classic dual-license cash cow.

2.4 The CalVer signal

Versioning by calendar date (`v26.6.9` = June 9, 2026) is itself a licensing instrument: the BUSL Change Date is defined per release, so CalVer makes the conversion schedule legible at a glance. A customer can see precisely when any given version becomes AGPL, which converts license anxiety into a predictable procurement calculation.

Chapter 3

Evidence-as-a-Service

3.1 The category

Process mining vendors sell *insight*: dashboards showing where a process deviates. wasm4pm can sell something categorically different: *evidence* — a cryptographically verifiable record that a process conformed (or did not) to its declared model at a point in time. The difference matters because insight is consumed by operations teams with discretionary budgets, while evidence is consumed by compliance, legal, and audit functions with *mandatory* budgets.

3.2 The mechanism

Three platform components compose into the evidence pipeline:

1. **Receipts.** Every run emits a BLAKE3 receipt binding input hashes to output hashes. Tampering with either side is detectable by anyone holding the receipt.
2. **TrueX envelopes.** Conformance admission records carrying an OCEL 2.0 batch hash, an equivalence class, and an expected path hash — a machine-checkable assertion that observed behavior matched a declared process model.
3. **The Supabase sync layer.** Receipts and envelopes flow to a central store with an offline queue and a dead-letter table, so evidence survives air-gapped and intermittently connected environments. The `live_verified` doctor status is itself a receipt: proof that the evidence pipeline was operational.

3.3 Who pays, and under which mandate

Sector	Mandate	What they buy
Pharma / medical devices	FDA 21 CFR Part 11, GxP	Proof that manufacturing and quality workflows followed validated procedures
Banking / payments	SOX, PSD2, model risk (SR 11-7)	Conformance evidence for transaction-processing and approval workflows
Healthcare providers	HIPAA audit controls	Verifiable access- and treatment-process logs
Public sector / defense	FedRAMP, supply-chain integrity	Receipts over procurement and deployment pipelines, generated on-premises
SaaS vendors	SOC 2 Type II	Continuous control-execution evidence instead of annual sampling

3.4 Why incumbents cannot trivially follow

Celonis, UiPath Process Mining, and SAP Signavio are cloud-resident: the customer ships event data to the vendor. That architecture is the wrong trust topology for evidence — the party generating the proof also custodies the data, which auditors discount. wasm4pm inverts it: the WASM core runs *inside the customer's boundary* (browser, on-prem Node, edge), data never leaves, and only hashes travel. The vendor never sees the event log yet the evidence remains independently verifiable. Retrofitting this onto a cloud-native incumbent is a re-architecture, not a feature.

3.5 Revenue shape

Evidence-as-a-Service is a subscription with compounding lock-in: the value of the receipt chain grows with its length, and migrating away means abandoning (or expensively re-anchoring) the accumulated history. Pricing per retained-and-verifiable receipt aligns revenue with exactly that accumulation.

Chapter 4

Edge Economics: Where WASM Changes the Cost Curve

4.1 The incumbent cost structure

Cloud process mining has a data-gravity problem. Event logs are large, regulated, and continuously produced; the incumbent model ships them to a central cluster, which means the vendor's cost of goods includes ingest bandwidth, storage, and compute for every customer event — and the customer's cost includes egress fees, DPAs, and data-residency review. A meaningful fraction of enterprise process mining deals die in the data-transfer security review, not on product merit.

4.2 The wasm4pm inversion

Because the entire algorithm catalog compiles to a self-contained WASM binary, the analysis moves to the data:

Browser. The 3.5 MB browser profile runs full discovery and conformance on an event log opened from the analyst's local disk. The marginal cost of serving an additional analyst is a CDN fetch — effectively zero COGS.

Edge runtimes. The ∼1.5 MB edge profile fits Cloudflare Workers and Fastly Compute limits, enabling conformance checks *in the request path*: an API gateway can verify that a transaction's event sequence conforms before forwarding it.

IoT / shop floor. The ∼500 KB–1 MB profiles run on gateways beside the production line, mining equipment event streams locally and emitting only receipts upstream — the only architecture that works for bandwidth-constrained or classified environments.

4.3 Business models this enables

1. **Conformance-as-middleware.** License the edge profile to API gateway and service mesh vendors as an embeddable policy engine: process conformance as a deploy-time or request-time check, priced per gateway node. The deterministic, side-effect-free WASM sandbox is precisely what infrastructure vendors will accept into their hot path.
2. **OEM embedding in machine tools and MES.** Manufacturing execution systems and equipment vendors embed the IoT profile and resell "self-auditing equipment." Per-device royalties; the `wasm4pm-compat` crates.io distribution is the natural channel.

3. **Zero-infrastructure SaaS.** A hosted product where the server stores only receipts and renders only dashboards, while all computation happens client-side. Gross margin approaches that of a static-site business while the product competes with compute-heavy incumbents.

4.4 The size gate as a commercial control point

The CI size-regression gate (baseline 3,551,895 bytes, ±5%) is not only engineering hygiene — it protects the edge value proposition. Every profile that fits inside a runtime's module limit is a market that stays addressable; uncontrolled binary growth silently closes those markets. The gate institutionalizes the constraint.

Chapter 5

Vertical Plays

The horizontal platform (Chapters 2–4) monetizes breadth. Verticals monetize *opinionation*: pre-built process models, conformance rule packs, and receipt schemas for one industry's workflows. Each vertical below is the same platform plus a domain pack — high-margin content layered on the existing engine.

5.1 CI/CD process assurance (the proven test case)

The repository already practices this on itself: proof gates run sixteen receipt-backed tests on every push, and the development constitution treats a model-versus-log mismatch as a defect. Productized, this is **pipeline conformance for software delivery**: mine the OTel traces of a CI/CD system into OCEL event logs, verify that every release followed the declared pipeline (no skipped stages, no out-of-band deploys), and emit receipts that map directly onto SOC 2 change-management controls and SLSA provenance requirements. Buyers: platform engineering and compliance at software-heavy enterprises. The DevOps toolchain integration surface (GitHub Actions, GitLab) is cheap to build because the CLI already exists.

5.2 Healthcare pathway conformance

Clinical pathways are declared process models; EHR audit trails are event logs. The browser profile means PHI never leaves the hospital network — the entire analysis runs inside the clinician portal. Receipts provide the HIPAA audit-control artifact. This vertical is gated on HL7/FHIR event adapters, not on algorithm work.

5.3 Financial transaction monitoring

KYC/AML processes are legally mandated sequences. The edge profile placed in the payment path can check conformance per transaction batch and emit TrueX envelopes as regulator-facing evidence. The determinism guarantee matters doubly here: model-risk-management frameworks (SR 11-7) require reproducible model behavior, which stochastic incumbent scoring cannot provide and wasm4pm provides by construction.

5.4 Manufacturing and Industry 4.0

OCEL 2.0 object-centric logs are the natural model for a shop floor, where one order touches many machines and one machine touches many orders. The IoT profile mines locally; receipts roll up to the MES. The sale is "digital thread with proof": every unit shipped carries a verifiable history of the process that produced it. This composes with the OEM model of Chapter 4 — the equipment vendor embeds, the manufacturer subscribes to the receipt store.

5.5 Sequencing

CI/CD assurance should lead: the domain knowledge is already in-house, the buyer is technical (short sales cycle), and every deal seeds the horizontal evidence platform that the regulated verticals — healthcare, finance, manufacturing — then justify with mandatory budgets and longer cycles.

Chapter 6

Ecosystem and Platform Strategy

6.1 The package surface as a funnel

The npm distribution is deliberately granular: thirteen `@wasm4pm/*` packages plus the bare `wasm4pm` WASM core, each independently installable. This is a funnel design. A developer who adopts only `@wasm4pm/contracts` for its receipt types, or only the WASM core for a one-off discovery task, is a qualified lead for the platform; the workspace-coherent versioning (every package at the same CalVer) makes the upgrade path from one package to the full stack frictionless.

6.2 The agent and MCP channel

Process mining's historical bottleneck is the analyst: someone must know which algorithm answers which question. Large-language-model agents dissolve that bottleneck, and wasm4pm is unusually well positioned for the agentic channel:

- Deterministic, schema-validated tools are exactly what agent frameworks need — a stochastic tool poisons an agent's reasoning, a deterministic one grounds it.
- Receipts give agent outputs provenance: when an agent claims "this process has a bottleneck at approval," the receipt chain shows which analysis, on which data, produced that claim. This is the antidote to hallucination in analytical workflows.
- MCP tooling exposes the catalog conversationally; every Claude or GPT deployment in an enterprise becomes a distribution endpoint that no incumbent sales force can match.

The commercial expression is metered tool-call pricing for hosted agent backends, plus the OEM license for agent-platform vendors who embed the WASM core directly.

6.3 Marketplace dynamics

Once the receipt format and the algorithm registry are stable public interfaces, third parties can sell into the ecosystem: domain-specific process model packs (Chapter 5), custom conformance rules, and adapters from proprietary event sources (SAP, Salesforce, Epic). The platform takes a revenue share and — more importantly — every third-party pack deepens the moat, because packs are written against the receipt schema and the OCEL contracts, not against a competitor's.

6.4 The standards position

OCEL 2.0 and XES are open standards; wasm4pm's strategic posture should be aggressive standards conformance plus one proprietary extension: the receipt and TrueX envelope schemas. Interoperate on data in, differentiate on evidence out. If the receipt format itself later becomes a de facto standard, first-mover verification infrastructure (the hosted receipt store, Chapter 3) is where the rent accrues — the Let's Encrypt lesson: the certificate can be free while the issuance and transparency infrastructure carries the strategic value.

Chapter 7

Risks, Honest Limitations, and Roadmap

7.1 Risks stated plainly

Commodity algorithms. pm4py is free, mature, and academically canonical. Any pitch resting on algorithm access loses. Mitigation: every commercial motion in this document prices receipts, deployment profiles, embedding rights, or domain packs — never algorithms.

BUSL backlash. License changes have cost projects community trust (HashiCorp's relicensing produced the OpenTofu fork). Mitigation: wasm4pm relicensed early, before a large external contributor base existed, and the AGPL conversion date is mechanically legible via CalVer.

Evidence without endorsement. Receipts only carry audit weight if auditors accept them. Until a Big Four firm or a regulator references receipt-based process evidence, Chapter 3 is a thesis, not a market. Mitigation: target SOC 2 auditors first — they are the most innovation-tolerant of the audit populations — and publish the verification spec openly.

Single-maintainer concentration. Enterprise procurement scores bus factor. Mitigation: the Fortune-5 readiness work (governance files, SECURITY.md, signed releases, provenance) addresses the paper-trail dimension; the organizational dimension requires hiring or a foundation arrangement and cannot be fixed in the repository.

Known technical gaps. CLI↔WASM trace correlation is deferred (spans appear disconnected in APM tools); stochastic algorithm seeds are fixed but not yet configurable. Both are documented in `docs/ENTERPRISE.md` — the discipline of shipping a Known Limitations section is itself a trust signal.

7.2 Sequenced roadmap

Horizon	Motion
Now–6 months	Land the dual-license funnel: publish all packages with provenance, convert BUSL-overage users to commercial licenses, ship the CI/CD assurance pack as the first vertical.
6–18 months	Stand up the hosted receipt store (Evidence-as-a-Service); pursue one design-partner auditor; release the MCP/agent tool surface as a metered channel.
18–36 months	OEM motion into gateway, MES, and agent-platform vendors; open the marketplace for third-party domain packs; push the receipt schema toward standardization.

7.3 Closing argument

Every durable software business owns a verb. Stripe owns *charge*; Docker owned *containerize*. The architecture documented here — deterministic WASM execution, BLAKE3 receipt chains, conformance envelopes, customer-boundary deployment — positions wasm4pm to own *prove*: not "our dashboard says the process is compliant," but "here is a receipt anyone can verify." Insight is a feature; evidence is a business.